2 Express each of the following in simplest form with a rational denominator.

WE10

(a) $\frac{1}{3-\sqrt{7}}$ (b) $\frac{4}{2+\sqrt{3}}$ (c) $\frac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$ (d) $\frac{2}{2\sqrt{7}+5}$

(e) $\frac{6+\sqrt{3}}{2\sqrt{3}-3}$ (f) $\frac{\sqrt{5}+1}{7-3\sqrt{5}}$ (g) $\frac{\sqrt{6}-1}{3\sqrt{2}+\sqrt{7}}$ (h) $\frac{\sqrt{2}-3}{2\sqrt{3}+\sqrt{6}}$

3 (a) $\frac{5}{\sqrt{2}}$ expressed with a rational denominator is:

A $\frac{\sqrt{10}}{2}$ B $\frac{2\sqrt{5}}{5}$ C $\frac{5\sqrt{2}}{\sqrt{2}}$ D $\frac{5\sqrt{2}}{2}$

(b) $\frac{4}{3\sqrt{3}}$ expressed with a rational denominator is:

A $\frac{4\sqrt{3}}{9}$ B $\frac{\sqrt{12}}{9}$ C $\frac{4\sqrt{3}}{3}$ D $\frac{4\sqrt{3}}{27}$

(c) $\frac{2\sqrt{2}}{3\sqrt{7}}$ expressed with a rational denominator is:

A $\frac{2\sqrt{6}}{21}$ B $\frac{2\sqrt{14}}{21}$ C $\frac{14\sqrt{2}}{3}$ D $\frac{2\sqrt{14}}{3}$

4 (a) The conjugate of $2\sqrt{3}-5$ is:

A $2\sqrt{3}-5$ B $2\sqrt{3}+5$ C $\sqrt{3}+5$ D $5\sqrt{3}+2$

(b) To rationalise the denominator of $\frac{4+\sqrt{7}}{2-3\sqrt{3}}$ it is necessary to multiply by:

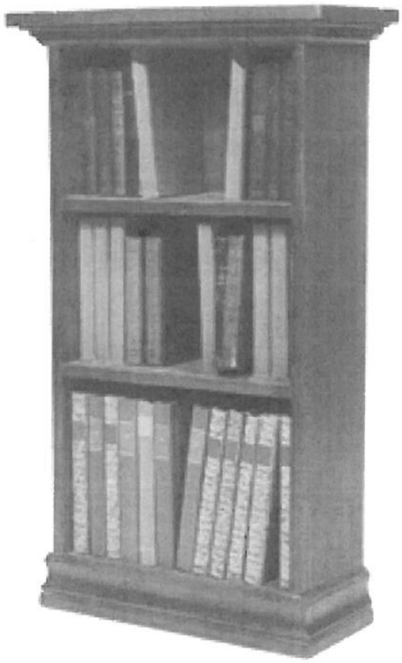
A $\frac{4-\sqrt{7}}{4-\sqrt{7}}$ B $\frac{1}{2+3\sqrt{3}}$ C $2+3\sqrt{3}$ D $\frac{2+3\sqrt{3}}{2+3\sqrt{3}}$

Understanding

5 Simplify each of the following. (Rationalise each denominator first.)

(a) $\frac{1}{\sqrt{3}+1}+\frac{1}{3-\sqrt{5}}$ (b) $\frac{1}{\sqrt{7}+1}+\frac{1}{3-\sqrt{6}}$

(c) $\frac{1}{\sqrt{6}+\sqrt{3}}-\frac{1}{4-\sqrt{11}}$ (d) $\frac{1}{\sqrt{5}+\sqrt{3}}-\frac{1}{\sqrt{7}-2}$



Reasoning

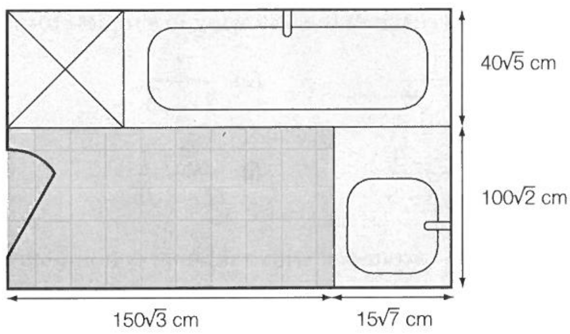
6 A piece of wood with length $5\sqrt{6}$ m is cut to make shelves for a bookcase with length $2\sqrt{2}$ m.

(a) Show $\frac{5\sqrt{6}}{2\sqrt{2}}$ as a fraction with a rational denominator.

(b) How many shelves can be cut from the wood?

(c) What is the exact length of wood left over?

7 The floor plan of a bathroom to be tiled is shown.



(a) Calculate the area to be tiled shaded orange.

(b) The square tiles chosen have side lengths of $10\sqrt{3}$ cm. How many tiles are required for the bathroom floor? State the answer as a simplified surd with a rational denominator.

(c) How many tiles would you order?

(d) Find the exact area of the entire bathroom in square centimetres.

8 If $\dfrac{3+2\sqrt{3}}{4-\sqrt{2}}a = \dfrac{4+\sqrt{2}}{5-6\sqrt{3}}$, find the value of a, expressing your answer with a rational denominator.

Open-ended

9 Find at least three fractions that are equivalent to $\dfrac{4\sqrt{2}}{\sqrt{3}}$, one of which must have a rational denominator.

10 For each of the following, find two fractions that the stated number is between, with a rational denominator formed from the original fraction.

(a) $\dfrac{1}{\sqrt{3}}$ (b) $\dfrac{3}{\sqrt{7}}$ (c) $\dfrac{2\sqrt{3}}{\sqrt{5}}$ (d) $\dfrac{2}{3\sqrt{2}}$

11 Find at least two expressions that $2+\sqrt{3}$ can be multiplied by to give a rational answer.

12 Sura is trying to rationalise the denominator of the fraction $\dfrac{\sqrt{2}-3}{2\sqrt{3}+\sqrt{6}}$. She is struggling to understand why the denominator can be rationalised by multiplying both the numerator and denominator by the denominator's conjugate $2\sqrt{3}-\sqrt{6}$. How could you explain this to Sura?

$$\frac{\sqrt{2}-3}{(2\sqrt{3}+\sqrt{6})} \times \frac{(2\sqrt{3}-\sqrt{6})}{(2\sqrt{3}-\sqrt{6})}$$